Lesson plan · for the teacher

Samples and uncertainty

Data science · Level 4 · Lesson 1

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; 60 numbered cards (or paper slips) and a bag; A phone or tablet with Darsi, for the lab

Learning goals

- See why samples vary.
- Report a margin of error.
- Ask who is missing.

Activity: Sixty cards

1. Warm-up (5 min): Show the two homework surveys and ask colleagues to vote on which teacher made a mistake.
2. Teach (8 min): Show twenty sample averages as a histogram and mark the true value.
3. Draw (15 min): Each pair draws samples of 5 and of 20 from the bag and adds their averages to two board plots.
4. Audit (7 min): Look at a recent school survey: how was it chosen, and who was missing?
5. Wrap-up (5 min): Ask two colleagues to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Which plot was tighter, and why?
- When is a gap between groups real?
- Who did our last survey miss?

Exit ticket

- Write a survey result as a range with a margin of error.

Try it at school tomorrow
Put 60 numbered cards in a bag; each group draws 10 and finds the average. Write every average on the board and ask why they differ.

Practice questions · for you

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1 **We want a smaller margin of error. What should we do?**

◯ Round the answer ◯ Ask only our friends ◯ Ask more pupils, chosen at random

2 **Two fair random samples of the same school can give different averages.**

◯ True ◯ False

3 **Print the average of the first 10 pupils, the average of the last 10, and the average of all 20.**

```
m = [40, 55, 60, 35, 70, 50, 45, 65, 55, 60, 50, 75, 40, 55, 65, 45, 60, 50, 70, 55]
first = m
print(round(sum(first) / 10, 1))
```

Write your program:

4 **How many times bigger must a sample be to halve the margin of error?**

◯ 2 times ◯ 10 times ◯ About 4 times

5 **A survey asked 500 pupils at the after-school club. What could go wrong? Choose all that apply.**

☐ 500 is too many to add up ☐ Pupils who leave early are missing

☐ Club pupils may differ from others

✎ **Which survey at your school might be missing a group? How would you reach them?**

Check your answers

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1 **Ask more pupils, chosen at random**

A bigger random sample wobbles less.

2 **True**

Samples vary by chance; neither has to be wrong.

3

```
m = [40, 55, 60, 35, 70, 50, 45, 65, 55, 60, 50, 75, 40, 55, 65, 45, 60, 50, 70, 55]
first = m[:10]
last = m[10:]
print(round(sum(first) / 10, 1), round(sum(last) / 10, 1), round(sum(m) / 20, 1))
```

The two samples give 53.5 and 56.5 minutes, wobbling either side of the full value, 55.0.

4 **About 4 times**

The margin shrinks with the square root of the sample size, so 4 times the pupils halves it.

5 **Pupils who leave early are missing; Club pupils may differ from others**

The sample is not random, so its size cannot fix who is missing.